Data Model & Database Design

1. Entity-Relationship Diagram

```mermaid erDiagram PERSONS ||--o| HOUSEHOLD_PROFILES : has PERSONS ||--o| CARDS : holds CARDS ||--o{ CARD_QUOTA_LEDGER : tracks LOTS ||--o{ LOT_CAP_GRANTS : issues LOT_CAP_GRANTS ||--o{ DISTRIBUTION_NODES : allocated_to DISTRIBUTION_NODES ||--o{ DISTRIBUTION_TRANSACTIONS : processes DISTRIBUTION_TRANSACTIONS ||--o{ TRANSACTION_STATE_HISTORY : tracks DISTRIBUTION_TRANSACTIONS ||--o| PROVISIONAL_TOKENS : generates 2. Data Dictionary (Key Tables) persons: id (UUID), first_name, last_name, phone_number (unique), dob, created_at. household_profiles: id, person_id (FK), family_count, yearly_income (DECIMAL(12,2)), address, city, state. cards: id (UUID), person_id (FK), status, issued_at. lots: id, name, grain_type, total_quantity (DECIMAL). lot_cap_grants: id, lot_id, node_id, max_quantity, expires_at, signature. card_quota_ledger: id, card_id, node_id, allocated_qty, consumed_qty, period_start. distribution_transactions: id, card_id, node_id, lot_id, quantity, state (ENUM), idempotency_key. transaction_state_history: id, txn_id, from_state, to_state, reason_code, actor_id, timestamp. provisional_tokens: id, txn_id, payload (JSON), signature, key_id, nonce, counter. failure_events: id, txn_id, reason_code, metadata (JSON), timestamp. audit_events: id, actor_id, action, resource_type, resource_id, details (JSON), timestamp.
```